FINDING POINTS OF FIBRE DIMENSION R AFTER A SURJECTIVE BASE CHANGE

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Allowed standard prerequisites (editorial).

- Dimension of fibres after base change (Morphisms, Lemma morphisms-lemma-dimension-fibre-after-base-change, canonical tag 02FY). cited at proof lines 809-810. Proves part (1): some point x' of X' over x has trdeg r over kappa(f'(x')).
- Points of a fibre product over a common point (Descent, Lemma descent-lemma-equiv-fibre-product, canonical tag 02KI). cited at proof lines 811-814. Supplies a point x" of X" = X' x_X X' mapping to both x'_1 and x'_2.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cited at proof lines 809-810: proves part (1).

Source reference. Original lemma 02KI: sources/stacks-chow/tags.txt, source lines 3373–3373. Cited at proof lines 811-814: supplies x" in X" mapping to both x'_1 and x'_2.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $r \geq 0$ be an integer. Let $g : S' \to S$ be a surjective morphism of schemes. Set $S'' = S' \times_S S'$ and let $f' : X' \to S'$ and $f'' : X'' \to S''$ be the base changes of f. Let $x \in X$ with $trdeg_{\kappa(f(x))}(\kappa(x)) = r$.*

(1) *There exists an $x' \in X'$ mapping to x with $trdeg_{\kappa(f'(x'))}(\kappa(x')) = r$.*
(2) *If $x'_1, x'_2 \in X'$ are both as in (1), then there exists an $x'' \in X''$ with $trdeg_{\kappa(f''(x''))}(\kappa(x'')) = r$ and $pr_i(x'') = x'_i$.*

Proof. Part (1) is Morphisms, Lemma 02FY. Let x'_1, x'_2 be as in (2). Then since $X'' = X' \times_X X'$ we see that there exists a $x'' \in X''$ mapping to both x'_1 and x'_2 (see for example Descent, Lemma 02KI). Denote $s'' \in S''$, $s'_i \in S'$, and $s \in S$ the images of x'', x'_i, and x. Denote $k = \kappa(s)$ and let $Z \subset X_k$ be the integral closed subscheme whose generic point is x. Then x'_i is a generic point of an irreducible component of $Z_{\kappa(s'_i)}$. Let $Z'' \subset Z_{\kappa(s'')}$ be an irreducible component containing x''. Denote $\xi'' \in Z''$ the generic point. Since $\xi'' \rightsquigarrow x''$ we see that ξ'' must also map to x'_i under

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the two projections. On the other hand, we see that $\mathrm{trdeg}_{\kappa(s'')}(\kappa(\xi'')) = r$ because it is a generic point of an irreducible component of the base change of Z. $\square$